

Niklas Rieken

Curriculum Vitae

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Personal Data

Date of Birth November 9, 1992
Place of Birth Krefeld, Germany

Education

May 2019 **Master of Science, RWTH Aachen, Computer Science**
Thesis: Network Pricing Games with Atomic Splittable Followers
Final Grade: 1.4
March 2017 **Bachelor of Science, RWTH Aachen, Computer Science**
Thesis: Alphabets and Machines as Structures and Their Decision Problems
Final Grade: 2.5
May 2012 **Abitur (German A-Levels), Michael-Ende-Gymnasium, Tönisvorst**
Advanced Courses: Math and Physics
Final Grade: 2.6

Work Experience

2019–today **PhD Student, Chair of Management Science, RWTH Aachen**
Researcher in Combinatorial Optimization and Algorithmic Game Theory.
Matchings, matroids, mechanism design, teaching courses in mathematical economics.
2017–2019 **Research Assistant, Chair of Operations Research, RWTH Aachen**
Developer for GCG (*generic column generation*, MIP solver).
Performance tests, extracting statistics, and collecting information on vast instance sets.
2014–2019 **Teaching Assistant, Chair of Computer Science 7 / TCS, RWTH Aachen**
TA for undergraduate courses in theoretical computer science.
Teaching exercise groups, grading assignments and exams.

Scientific Publications & Preprints

ANRW 2025 **Using Explicit (Host-to-Host) Flow Measurements for Network Tomography**
with Ike Kunze, Constantin Sander, Alexander Ruhrmann, and Klaus Wehrle
TEAC 2025 **Faster Dynamic Auctions via Polymatroid Sum**
with Katharina Eickhoff, Meike Neuwohner, Britta Peis, Laura Vargas Koch, and László A. Végh
Networks 2024 **A flow-based ascending auction to compute buyer-optimal Walrasian prices**
with Katharina Eickhoff, S. Thomas McCormick, Britta Peis, and Laura Vargas Koch
WINE 2023 **Faster Ascending Auctions via Polymatroid Sum**
with Katharina Eickhoff, Britta Peis, Laura Vargas Koch, and László A. Végh
MAPSP 2022 **A Primal-Dual and Primal-Greedy Approximation Framework for Weighted Covering Problems**
with Britta Peis, José Verschae, and Andreas Wierz
(Preprint) **A Simplified Analysis of the Ascending Auction to Sell a Matroid Base**
with Britta Peis

Scientific Talks

July 3, 2025 **A Self-Reflecting Greedy Algorithm for Submodular Cover**
10th Gerhard Woeginger Research Colloquium, RWTH Aachen
May 27, 2025 **Combinatorial Optimization in Auction Design**
Seminar on Microeconomics, RWTH Aachen

November 28, 2023	Selling Bases of a Matroid ORM PhD Seminar, RWTH Aachen
August 30, 2023	An Ascending Auction for Computing Minimal Walrasian prices via the Matroid Partitioning Algorithm OR 2023, Universität Hamburg
June 15, 2023	Selling Bases of a Matroid 13th Day on Computational Game Theory, Vrije Universiteit Amsterdam
January 18, 2023	Computing Walrasian Prices via Matroid Partitioning Santiago Workshop on Combinatorial Optimization, Universidad de Chile
July 6, 2022	Computing Buyer-Optimal Walrasian Prices in Multi-Unit Matching Markets via a Sequence of Max-Flow Computations Seminario AGCO, Universidad de Chile

Awards

November 9, 2024	Prize for outstanding teaching activities Faculty 8 of Economics, RWTH Aachen University
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Coursework

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| <ul style="list-style-type: none"> ○ Complexity Theory ○ Recursion Theory ○ Mathematical Logic I + II ○ Logic and Games ○ Applied Automata Theory ○ Infinite Computations | <ul style="list-style-type: none"> ○ Convex Optimization ○ Operations Research I + II ○ Optimization with Modelling Languages ○ Methods and Applications of Optimization ○ Discrete Differential Geometry ○ Satisfiability Checking |
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Skills

Operating Systems	GNU/Linux
Programming	Python, Bash, C/C++
Applications	L^AT_EX(+TikZ), Gurobi

Languages

German	native
English	proficient
Spanish	basics

Professional Interests

- Mathematical Optimization
- Theory of Formal Languages
- Algorithmic Game Theory

Personal Interests

- Ultimate and Table Tennis
- Guitar and Ukulele
- Reading, Movies, and Music
- Cooking, Board games